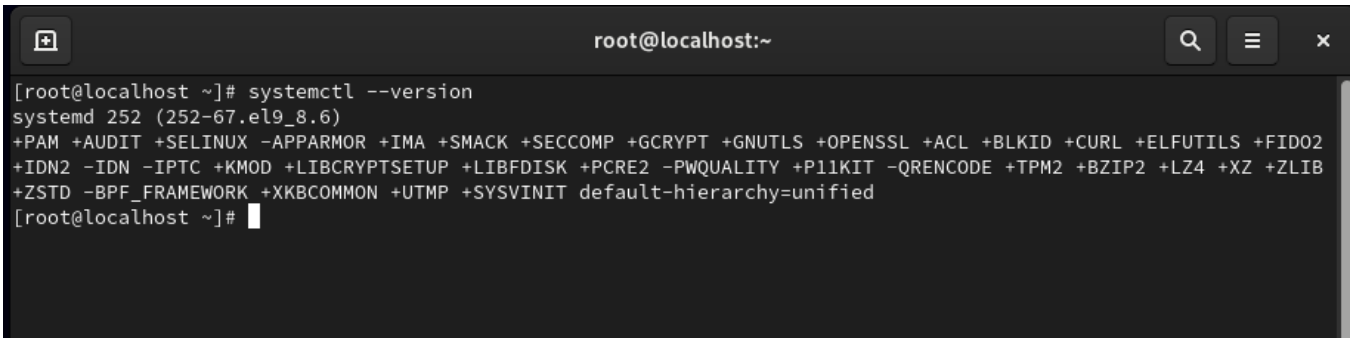
COMPTE RENDU DE TRAVAUX PRATIQUES

Administration Unix – 4IIR

TP n°3 : Configuration des cibles et gestion des services par Systemd

Atelier 1 : Bases de systemd et systemctl

1. Vérifier si le système utilise Systemd



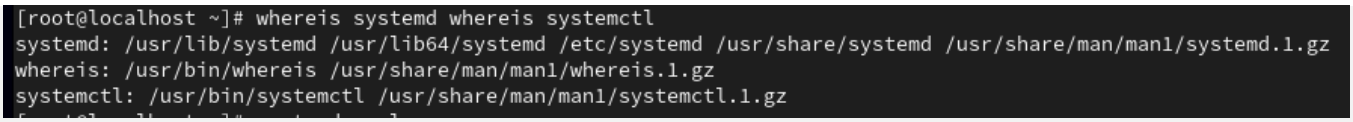
Capture d'écran 1 : Résultat de systemctl --version

2. Vérifier le gestionnaire de démarrage (PID 1)



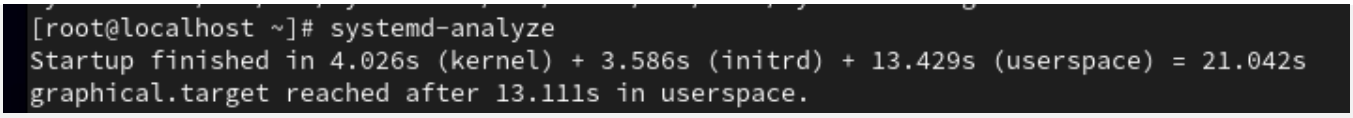
Capture d'écran 2 : ps aux | head -2 (PID 1 = systemd)

3. Emplacement des binaires systemd et systemctl



Capture d'écran 3 : whereis systemd et whereis systemctl

4. Temps de démarrage du système



Capture d'écran 4 : Résultat de systemd-analyze

5. Temps pris par chaque processus au démarrage

```
[root@localhost ~]# systemd-analyze blame
7.635s kdump.service
7.574s plymouth-quit-wait.service
3.989s sys-module-fuse.device
3.716s dev-ttyS1.device
3.716s sys-devices-platform-serial8250-tty-ttyS1.device
3.711s sys-devices-platform-serial8250-tty-ttyS3.device
3.711s dev-ttyS3.device
3.710s dev-ttyS2.device
3.710s sys-devices-platform-serial8250-tty-ttyS2.device
3.709s dev-ttyS0.device
3.709s sys-devices-platform-serial8250-tty-ttyS0.device
3.614s sys-module-configfs.device
3.436s sys-subsystem-net-devices-ens160.device
3.436s sys-devices-pci0000:00-0000:00:15.0-0000:03:00.0-net-ens160.device
3.402s dev-nvme0n1p3.device
3.402s dev-disk-by\x2dpath-pci\x2d0000:0b:00.0\x2dnvme\x2d1\x2dpart3.device
3.402s sys-devices-pci0000:00-0000:00:16.0-0000:0b:00.0-nvme-nvme0-nvme0n1-nvme0n1p3.device
3.402s dev-disk-by\x2did-nvme\x2deui.716ade64119eb74f000c2969b09af896\x2dpart3.device
```

Capture d'écran 5 : systemd-analyze blame

6. Chaîne critique du démarrage

```
[root@localhost ~]# systemd-analyze critical-chain
The time when unit became active or started is printed after the "@" character.
The time the unit took to start is printed after the "+" character.

graphical.target @13.111s
└─multi-user.target @13.111s
  └─plymouth-quit-wait.service @5.536s +7.574s
    └─systemd-user-sessions.service @5.457s +61ms
      └─network.target @5.424s
        └─wpa_supplicant.service @8.922s +37ms
          └─basic.target @3.586s
            └─dbus-broker.service @3.538s +44ms
              └─dbus.socket @3.527s
                └─sysinit.target @3.523s
                  └─systemd-update-utmp.service @3.513s +9ms
                    └─auditd.service @3.456s +51ms
                      └─systemd-tmpfiles-setup.service @3.309s +141ms
                        └─local-fs.target @3.290s
                          └─run-credentials-systemd\x2dtmpfiles\x2dsetup.service.mount @3.315s
                            └─local-fs-pre.target @3.188s
                              └─systemd-tmpfiles-setup-dev.service @656ms +95ms
                                └─kmod-static-nodes.service @536ms +62ms
                                  └─systemd-journald.socket
                                    └─system.slice
                                      └─-.slice
[root@localhost ~]#
```

Capture d'écran 6 : systemd-analyze critical-chain

Atelier 2 : Configurer les cibles systemd

1. Afficher la cible par défaut

```
[root@localhost ~]# systemctl get-default
graphical.target
```

Capture d'écran 7 : systemctl get-default

2. Afficher les cibles actuellement chargées

```
[root@localhost ~]# systemctl get-default
graphical.target
[root@localhost ~]# systemctl list-units --type target
  UNIT                   LOAD   ACTIVE SUB    DESCRIPTION
  basic.target           loaded active active Basic System
  cryptsetup.target      loaded active active Local Encrypted Volumes
  getty.target           loaded active active Login Prompts
  graphical.target       loaded active active Graphical Interface
  integritysetup.target  loaded active active Local Integrity Protected Volumes
  local-fs-pre.target    loaded active active Preparation for Local File Systems
  local-fs.target        loaded active active Local File Systems
  multi-user.target      loaded active active Multi-User System
  network-online.target  loaded active active Network is Online
  network-pre.target     loaded active active Preparation for Network
  network.target         loaded active active Network
  nss-user-lookup.target loaded active active User and Group Name Lookups
  paths.target           loaded active active Path Units
  remote-fs.target       loaded active active Remote File Systems
  slices.target          loaded active active Slice Units
  sockets.target         loaded active active Socket Units
  sound.target           loaded active active Sound Card
  sshd-keygen.target     loaded active active sshd-keygen.target
  swap.target            loaded active active Swaps
  sysinit.target         loaded active active System Initialization
  timers.target          loaded active active Timer Units
  veritysetup.target     loaded active active Local Verity Protected Volumes

LOAD   = Reflects whether the unit definition was properly loaded.
ACTIVE = The high-level unit activation state, i.e. generalization of SUB.
SUB    = The low-level unit activation state, values depend on unit type.
22 loaded units listed. Pass --all to see loaded but inactive units, too.
To show all installed unit files use 'systemctl list-unit-files'.
[root@localhost ~]# S
```

Capture d'écran 8 : systemctl list-units --type target

3. Modifier la cible par défaut

```
[root@localhost ~]# systemctl set-default multi-user.target
Removed "/etc/systemd/system/default.target".
Created symlink /etc/systemd/system/default.target → /usr/lib/systemd/system/multi-user.target.
```

Capture d'écran 9 : systemctl set-default multi-user.target

4. Modifier la cible actuelle (isolate)

```
Red Hat Enterprise Linux 9.8 (Plow)
Kernel 5.14.0-687.54.1.el9_8.x86_64 on an x86_64

Activate the web console with: systemctl enable --now cockpit.socket

localhost login: emsi
Password:
Last login: Mon Nov  2 12:14:43 on tty2
[emsi@localhost ~]$ _
```

Capture d'écran 10 : systemctl isolate multi-user.target

5. Lister les unités défaillantes

```
[emsi@localhost ~]$ systemctl --failed
  UNIT LOAD ACTIVE SUB DESCRIPTION
0 loaded units listed.
[emsi@localhost ~]$
```

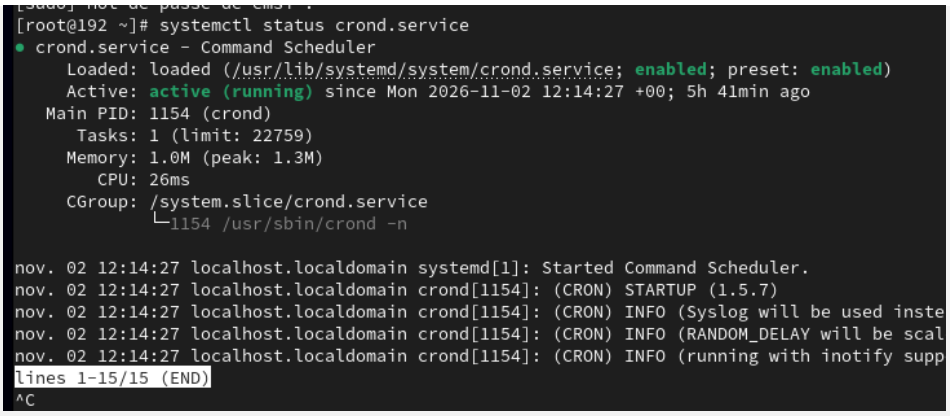
Capture d'écran 11 : systemctl --failed

6. Vérifier si un service est activé

```
[emsi@localhost ~]$ systemctl is-enabled sshd.service
enabled
```

Capture d'écran 12 : systemctl is-enabled sshd.service

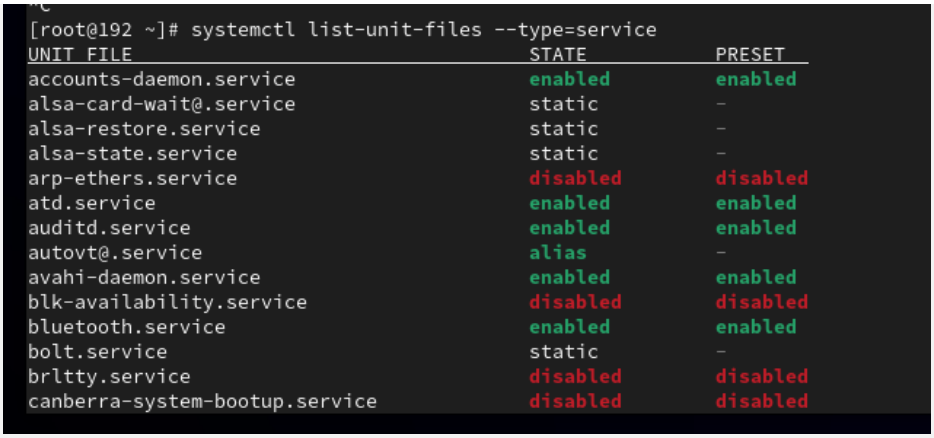
7. Vérifier le statut d'un service



Capture d'écran 13 : systemctl status crond.service

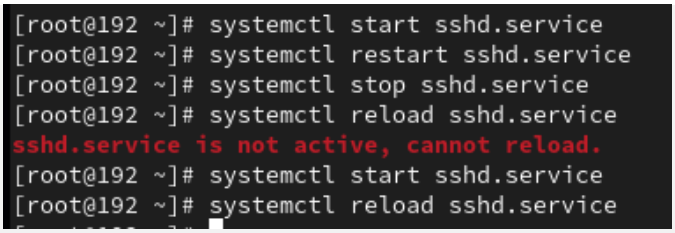
Atelier 3 : Gestion de services par systemctl

1. Lister tous les services



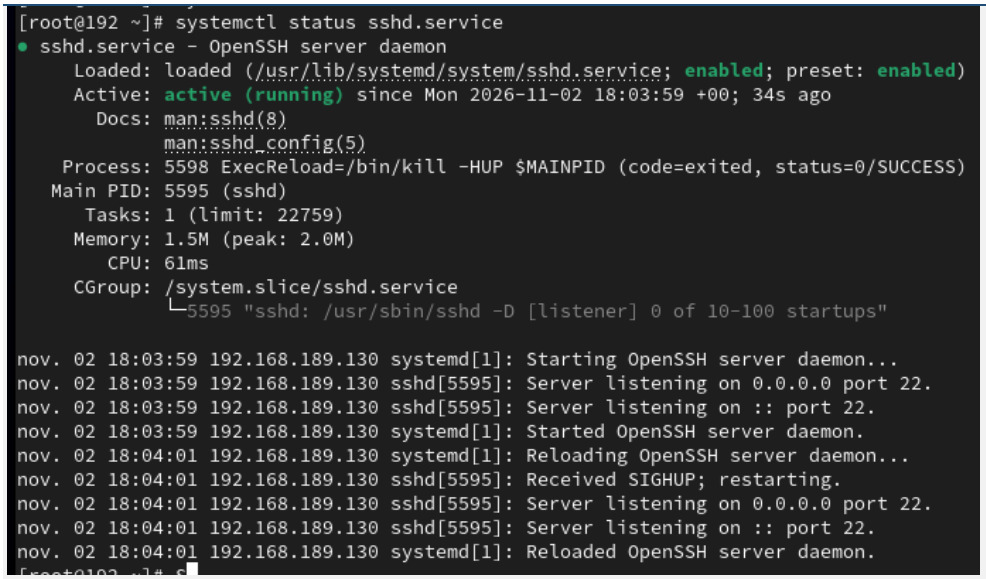
Capture d'écran 14 : systemctl list-unit-files --type=service

2. Démarrer / Redémarrer / Arrêter / Recharger le service SSH



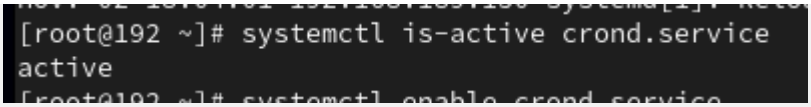
Capture d'écran 15 : Commandes start / restart / stop / reload sur sshd

3. Statut du service sshd



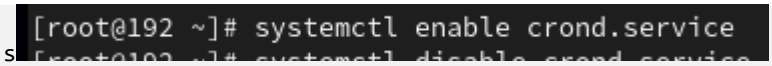
Capture d'écran 16 : systemctl status sshd.service

4. Vérifier si un service est actif



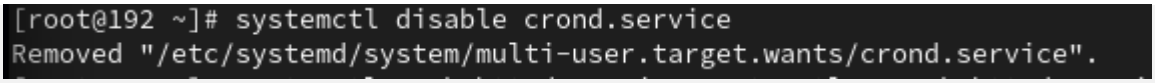
Capture d'écran 17 : systemctl is-active crond.service

5. Activer le démarrage automatique d'un service



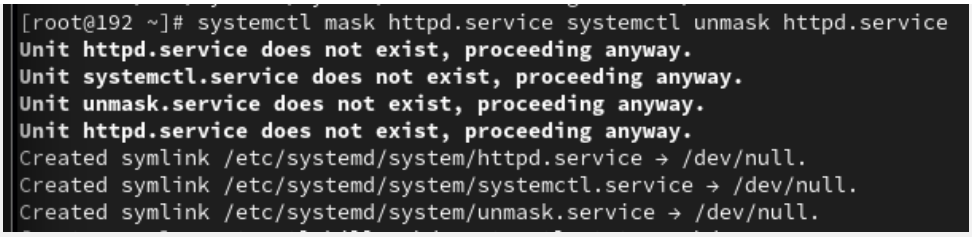
Capture d'écran 18 : systemctl enable crond.service

6. Désactiver le démarrage automatique d'un service



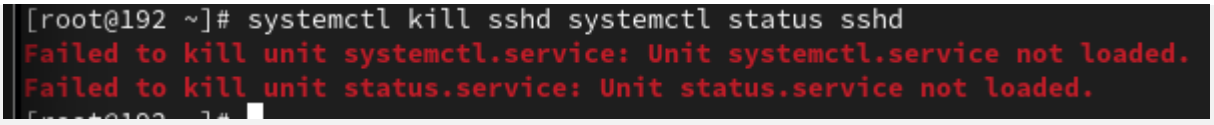
Capture d'écran 19 : systemctl disable crond.service

7. Masquer / Démasquer un service



Capture d'écran 20 : systemctl mask / unmask httpd.service

8. Envoyer un signal d'arrêt à un service



Capture d'écran 21 : systemctl kill sshd + status

Atelier 4 : Journalisation avec journalctl

1. Journaux depuis le dernier démarrage

```
[root@192 ~]# journalctl -b
nov. 02 12:14:19 localhost kernel: Linux version 5.14.0-687.54.1.el9_8.x86_64 (mockbuild@x86-64-04.build.eng.rdu2>
nov. 02 12:14:19 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Linu>
nov. 02 12:14:19 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-687.54.1.el9_8.x86_64 root=>
nov. 02 12:14:19 localhost kernel: Disabled fast string operations
nov. 02 12:14:19 localhost kernel: BIOS-provided physical RAM map:
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x000000000009ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x00000000000c0000-0x00000000000fffff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x000000000d97ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x000000000d980000-0x000000000d980fff] ACPI data
```

Capture d'écran 22 : journalctl -b

2. Logs en temps réel (follow)

```
[root@192 ~]# journalctl -f
nov. 02 18:05:34 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:34 192.168.189.130 systemd-rc-local-generator[5623]: /etc/rc.d/rc.local is not marked executable, sk
ipping.
nov. 02 18:05:41 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:42 192.168.189.130 systemd-rc-local-generator[5658]: /etc/rc.d/rc.local is not marked executable, sk
ipping.
nov. 02 18:05:49 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:49 192.168.189.130 systemd-rc-local-generator[5693]: /etc/rc.d/rc.local is not marked executable, sk
ipping.
nov. 02 18:05:57 192.168.189.130 sshd[5595]: Received signal 15; terminating.
nov. 02 18:05:57 192.168.189.130 systemd[1]: sshd.service: Sent signal SIGTERM to main process 5595 (sshd) on clie
nt request.
nov. 02 18:05:57 192.168.189.130 systemd[1]: sshd.service: Deactivated successfully.
nov. 02 18:08:17 192.168.189.130 gsd-color[4911]: unable to get EDID for xrandr-Virtual-1: unable to get EDID for
output
```

Capture d'écran 23 : journalctl -f

3. Logs d'un service spécifique (SSH)

```
[root@192 ~]# journalctl -u sshd.service
nov. 02 12:14:27 localhost.localdomain systemd[1]: Starting OpenSSH server daemon...
nov. 02 12:14:28 localhost.localdomain sshd[1140]: Server listening on 0.0.0.0 port 22.
nov. 02 12:14:28 localhost.localdomain systemd[1]: Started OpenSSH server daemon.
nov. 02 12:14:28 localhost.localdomain sshd[1140]: Server listening on :: port 22.
nov. 02 18:03:32 192.168.189.130 systemd[1]: Stopping OpenSSH server daemon...
nov. 02 18:03:32 192.168.189.130 sshd[1140]: Received signal 15; terminating.
nov. 02 18:03:32 192.168.189.130 systemd[1]: sshd.service: Deactivated successfully.
nov. 02 18:03:32 192.168.189.130 systemd[1]: Stopped OpenSSH server daemon.
nov. 02 18:03:32 192.168.189.130 systemd[1]: Starting OpenSSH server daemon...
nov. 02 18:03:32 192.168.189.130 sshd[5586]: Server listening on 0.0.0.0 port 22.
nov. 02 18:03:32 192.168.189.130 sshd[5586]: Server listening on :: port 22.
nov. 02 18:03:32 192.168.189.130 systemd[1]: Started OpenSSH server daemon.
nov. 02 18:03:47 192.168.189.130 sshd[5586]: Received signal 15; terminating.
nov. 02 18:03:47 192.168.189.130 systemd[1]: Stopping OpenSSH server daemon...
nov. 02 18:03:47 192.168.189.130 systemd[1]: sshd.service: Deactivated successfully.
```

Capture d'écran 24 : journalctl -u sshd.service

4. Logs en temps réel pour un service

```
[root@192 ~]# journalctl -u sshd.service -f
nov. 02 18:03:59 192.168.189.130 sshd[5595]: Server listening on :: port 22.
nov. 02 18:03:59 192.168.189.130 systemd[1]: Started OpenSSH server daemon.
nov. 02 18:04:01 192.168.189.130 systemd[1]: Reloading OpenSSH server daemon...
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Received SIGHUP; restarting.
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Server listening on 0.0.0.0 port 22.
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Server listening on :: port 22.
nov. 02 18:04:01 192.168.189.130 systemd[1]: Reloaded OpenSSH server daemon.
nov. 02 18:05:57 192.168.189.130 sshd[5595]: Received signal 15; terminating.
nov. 02 18:05:57 192.168.189.130 systemd[1]: sshd.service: Sent signal SIGTERM to main process 5595 (sshd)
```

Capture d'écran 25 : journalctl -u sshd.service -f

5. Logs par niveau de priorité

```
[root@192 ~]# journalctl -p err
nov. 02 12:14:24 localhost kernel: piix4_smbus 0000:00:07.3: SMBus base address uninitialized - upgrade BIOS or u
nov. 02 12:14:26 localhost alsactl[957]: alsa-lib main.c:1804:(snd_use_case_mgr_open) [error.ucm] failed to impor
nov. 02 12:14:42 localhost.localdomain gdm-password][2662]: gkr-pam: unable to locate daemon control file
nov. 02 12:14:49 localhost.localdomain gdm-wayland-session[1296]: GLib: Source ID 2 was not found when attempting
nov. 02 12:14:50 localhost.localdomain gdm-launch-environment][1170]: GLib-GObject: g_object_unref: assertion 'G_
nov. 02 12:26:50 localhost.localdomain gdm-password][3592]: gkr-pam: the password for the login keyring was inval
nov. 02 12:36:50 localhost.localdomain gdm-wayland-session[2697]: GLib: Source ID 2 was not found when attempting
nov. 02 17:46:12 192.168.189.130 gdm-password][4629]: gkr-pam: unable to locate daemon control file
nov. 02 17:46:15 192.168.189.130 systemd[4642]: Failed to start Application launched by gnome-session-binary.
nov. 02 17:46:15 192.168.189.130 systemd[4642]: Failed to start Application launched by gnome-session-binary.
nov. 02 17:46:24 192.168.189.130 gdm-wayland-session[4089]: GLib: Source ID 2 was not found when attempting to re
nov. 02 17:46:24 192.168.189.130 gdm-launch-environment][3961]: GLib-GObject: g_object_unref: assertion 'G_IS_OBJ

[root@192 ~]# journalctl -p warning
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'cpu cycles' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'instructions' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'bus cycles' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'cache references' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'cache misses' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'branch instructions' unavailable
nov. 02 12:14:19 localhost kernel: core: CPUID marked event: 'branch misses' unavailable
nov. 02 12:14:19 localhost kernel: device-mapper: core: CONFIG_IMA_DISABLE_HTABLE is disabled. Duplicate I
nov. 02 12:14:19 localhost systemd[1]: sys-module-fuse.device: Failed to enqueue SYSTEMD_WANTS= job, ignor
nov. 02 12:14:20 localhost kernel: Must use shrinker_alloc() to dynamically allocate the shrinker

[root@192 ~]# journalctl -p info
nov. 02 12:14:19 localhost kernel: Linux version 5.14.0-687.54.1.el9_8.x86_64 (mockbuild@x86-64-04.build.eng.rdu2>
nov. 02 12:14:19 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Linu>
nov. 02 12:14:19 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-687.54.1.el9_8.x86_64 root=>
nov. 02 12:14:19 localhost kernel: Disabled fast string operations
nov. 02 12:14:19 localhost kernel: BIOS-provided physical RAM map:
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x000000000009ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x00000000000c0000-0x00000000000fffff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x000000000d97ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x000000000d980000-0x000000000d980fff] ACPI data
```

Capture d'écran 26 : journalctl -p err / warning / info

6. Logs depuis une date/heure

```
[root@192 ~]# journalctl --since "today"
nov. 02 12:14:19 localhost kernel: Linux version 5.14.0-687.54.1.el9_8.x86_64 (mockbuild@x86-64-04.build.eng.rdu
nov. 02 12:14:19 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Lin
nov. 02 12:14:19 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-687.54.1.el9_8.x86_64 root
nov. 02 12:14:19 localhost kernel: Disabled fast string operations
nov. 02 12:14:19 localhost kernel: BIOS-provided physical RAM map:
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x000000000009ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x00000000000c0000-0x00000000000fffff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x000000000d97ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x000000000d980000-0x000000000d980fff] ACPI data
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x000000000d981000-0x000000000d989fff] usable

[root@192 ~]# journalctl --since "14:00"
nov. 02 17:46:01 localhost.localdomain chronyd[970]: Forward time jump detected!
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.6579] manager: sleep: wake reque>
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.6621] device (ens160): state cha>
nov. 02 17:46:01 localhost.localdomain kernel: vmxnet3 0000:03:00.0 ens160: intr type 3, mode 0, 3 vectors alloca>
nov. 02 17:46:01 localhost.localdomain kernel: vmxnet3 0000:03:00.0 ens160: NIC Link is Up 10000 Mbps
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.7653] manager: NetworkManager st>
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.7665] audit: op="sleep-control" >
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.7765] device (ens160): carrier: >
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.7815] device (ens160): state cha>
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.7892] policy: auto-activating co>
nov. 02 17:46:01 localhost.localdomain NetworkManager[1129]: <info>  [1793641561.8010] device (ens160): Activatio>

[root@192 ~]# journalctl --since "2024-01-15 09:00:00"
nov. 02 12:14:19 localhost kernel: Linux version 5.14.0-687.54.1.el9_8.x86_64 (mockbuild@x86-64-04.build.eng.rdu2>
nov. 02 12:14:19 localhost kernel: The list of certified hardware and cloud instances for Red Hat Enterprise Linu>
nov. 02 12:14:19 localhost kernel: Command line: BOOT_IMAGE=(hd0,gpt2)/vmlinuz-5.14.0-687.54.1.el9_8.x86_64 root=>
nov. 02 12:14:19 localhost kernel: Disabled fast string operations
nov. 02 12:14:19 localhost kernel: BIOS-provided physical RAM map:
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000000000-0x0000000000000fff] ACPI NVS
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000001000-0x0000000000001fff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000002000-0x000000000009ffff] usable
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x00000000000c0000-0x00000000000fffff] reserved
nov. 02 12:14:19 localhost kernel: BIOS-e820: [mem 0x0000000000100000-0x000000000d97ffff] usable
```

Capture d'écran 27 : journalctl --since ...

7. Nombre limité d'entrées

```
[root@192 ~]# journalctl -n 20
nov. 02 18:03:59 192.168.189.130 systemd[1]: Starting OpenSSH server daemon...
nov. 02 18:03:59 192.168.189.130 sshd[5595]: Server listening on 0.0.0.0 port 22.
nov. 02 18:03:59 192.168.189.130 sshd[5595]: Server listening on :: port 22.
nov. 02 18:03:59 192.168.189.130 systemd[1]: Started OpenSSH server daemon.
nov. 02 18:04:01 192.168.189.130 systemd[1]: Reloading OpenSSH server daemon...
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Received SIGHUP; restarting.
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Server listening on 0.0.0.0 port 22.
nov. 02 18:04:01 192.168.189.130 sshd[5595]: Server listening on :: port 22.
nov. 02 18:04:01 192.168.189.130 systemd[1]: Reloaded OpenSSH server daemon.
nov. 02 18:05:34 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:34 192.168.189.130 systemd-rc-local-generator[5623]: /etc/rc.d/rc.local is not marked executable, s>
nov. 02 18:05:41 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:42 192.168.189.130 systemd-rc-local-generator[5658]: /etc/rc.d/rc.local is not marked executable, s>
nov. 02 18:05:49 192.168.189.130 systemd[1]: Reloading.
nov. 02 18:05:49 192.168.189.130 systemd-rc-local-generator[5693]: /etc/rc.d/rc.local is not marked executable, s>
nov. 02 18:05:57 192.168.189.130 sshd[5595]: Received signal 15; terminating.
nov. 02 18:05:57 192.168.189.130 systemd[1]: sshd.service: Sent signal SIGTERM to main process 5595 (sshd) on cli>
nov. 02 18:05:57 192.168.189.130 systemd[1]: sshd.service: Deactivated successfully.
nov. 02 18:08:17 192.168.189.130 gsd-color[4911]: unable to get EDID for xrandr-Virtual-1: unable to get EDID for>
nov. 02 18:12:46 192.168.189.130 gnome-shell[4729]: libinput error: event2  - VirtualPS/2 VMware VMMouse: client >
lines 1-20/20 (END)
```

Capture d'écran 28 : journalctl -n 20

8. Formats de sortie avancés

```
[root@192 ~]# journalctl -u sshd.service -o json-pretty
{
        "_SYSTEMD_SLICE" : "-.slice",
        "_UID" : "0",
        "MESSAGE_ID" : "7d4958e842da4a758f6c1cdc7b36dcc5",
        "__MONOTONIC_TIMESTAMP" : "13063650",
        "_BOOT_ID" : "247b5101b5a84799816fc9bc63df44d5",
        "MESSAGE" : "Starting OpenSSH server daemon...",
        "INVOCATION_ID" : "6d2db2a1291b4e5e84d983806a2dfaec",
        "_TRANSPORT" : "journal",
        "JOB_TYPE" : "start",
        "__REALTIME_TIMESTAMP" : "1793621667869118",
        "_SOURCE_REALTIME_TIMESTAMP" : "1793621667864951",
        "_CAP_EFFECTIVE" : "1ffffffffff",
        "_SELINUX_CONTEXT" : "system_u:system_r:init_t:s0",
        "CODE_FUNC" : "job_emit_start_message",
        "SYSLOG_IDENTIFIER" : "systemd",
        "_COMM" : "systemd",
        "JOB_ID" : "264",
        "_HOSTNAME" : "localhost.localdomain"
```

```
[root@192 ~]# journalctl -u sshd.service -o cat
Starting OpenSSH server daemon...
Server listening on 0.0.0.0 port 22.
Started OpenSSH server daemon.
Server listening on :: port 22.
Stopping OpenSSH server daemon...
Received signal 15; terminating.
sshd.service: Deactivated successfully.
Stopped OpenSSH server daemon.
Starting OpenSSH server daemon...
Server listening on 0.0.0.0 port 22.
Server listening on :: port 22.
Started OpenSSH server daemon.
Received signal 15; terminating.
Stopping OpenSSH server daemon...
```

```
[root@192 ~]# journalctl -u sshd.service -o cat
Starting OpenSSH server daemon...
Server listening on 0.0.0.0 port 22.
Started OpenSSH server daemon.
Server listening on :: port 22.
Stopping OpenSSH server daemon...
Received signal 15; terminating.
sshd.service: Deactivated successfully.
Stopped OpenSSH server daemon.
Starting OpenSSH server daemon...
Server listening on 0.0.0.0 port 22
```

```
[root@192 ~]# journalctl -u sshd.service -o verbose
Mon 2026-11-02 12:14:27.864951 +00 [s=f601479ca5cc4b69a1aa41fad4b42dde;i=8df;b=247b5101b5a84799816fc9bc63df44d5;m>
    _BOOT_ID=247b5101b5a84799816fc9bc63df44d5
    _MACHINE_ID=9a11b742e7624c86876f8b95f06ad5fe
    PRIORITY=6
    SYSLOG_FACILITY=3
    SYSLOG_IDENTIFIER=systemd
    _UID=0
    _GID=0
    _TRANSPORT=journal
```

Capture d'écran 29 : journalctl formats json-pretty et cat

Atelier Supplémentaire : Création d'un service personnalisé

1. Création du script

```
[root@192 ~]# cat> custom_service.sh
#!/bin/bash
echo "Bonjour depuis le service personnalisé de systemd" >
/tmp/custom_service_output.txt
```

Capture d'écran 30 : Contenu du script custom_service.sh

2. Fichier unit systemd

```
[root@192 system]# cat> /etc/systemd/system/custom_service.service
[Unit]
Description=Service personnalisé de test

[Service]
ExecStart=/bin/bash /root/custom_service.sh
Type=oneshot
RemainAfterExit=yes

[Install]
WantedBy=multi-user.target
```

Capture d'écran 31 : Fichier /etc/systemd/system/custom_service.service

3. Activation et démarrage du service

```
[root@192 system]# systemctl daemon-reload
systemctl start custom_service.service
systemctl status custom_service.service
● custom_service.service - Service personnalisé de test
     Loaded: loaded (/etc/systemd/system/custom_service.service; enabled; preset: disabled)
     Active: active (exited) since Mon 2026-11-02 18:24:54 +00; 25ms ago
    Process: 5923 ExecStart=/bin/bash /root/custom_service.sh (code=exited, status=0/SUCCESS)
   Main PID: 5923 (code=exited, status=0/SUCCESS)
        CPU: 6ms

nov. 02 18:24:54 192.168.189.130 systemd[1]: Starting Service personnalisé de test...
nov. 02 18:24:54 192.168.189.130 systemd[1]: Finished Service personnalisé de test.
```

Capture d'écran 32 : daemon-reload + enable + start du service personnalisé

4. Vérification du fonctionnement

```
[root@192 system]# cat /tmp/custom_service_output.txt
systemctl status custom_service.service
Bonjour depuis le service personnalisé de systemd
● custom_service.service - Service personnalisé de test
     Loaded: loaded (/etc/systemd/system/custom_service.service; enabled; preset: disabled)
     Active: active (exited) since Mon 2026-11-02 18:24:54 +00; 1min 30s ago
    Process: 5923 ExecStart=/bin/bash /root/custom_service.sh (code=exited, status=0/SUCCES
   Main PID: 5923 (code=exited, status=0/SUCCESS)
        CPU: 6ms

nov. 02 18:24:54 192.168.189.130 systemd[1]: Starting Service personnalisé de test...
nov. 02 18:24:54 192.168.189.130 systemd[1]: Finished Service personnalisé de test.
```

Capture d'écran 33 : Vérification du fichier de sortie et statut du service